With vs Without Class-Imbalance Handling: Metric Differences

BRFSS 2015–2024 diabetes prediction · P1–P4 integrated comparison · test set 459,434 rows · threshold 0.5

Reading this table. ***Without*** = the **No Balancing** baseline (model trained on the imbalanced data as-is). ***With*** = each of the 11 resampling techniques. Δ = technique − baseline *for the same classifier*, so every delta is a like-for-like comparison. Green = higher than baseline, red = lower. A negative Δ on accuracy/precision is the *expected* trade-off for higher recall, not a failure — see takeaways below. † SVM-SMOTE and SMOTE-N were trained on a shared 50,000-row subsample.

Source: `P4_Integration_Evaluation/results/combined_results.csv` (36 rows; every value re-verified from saved prediction files, gate 1: 82/82).

Table 1. Logistic Regression — baseline vs each technique (deltas)

P	Technique	F1	ΔF1	Recall	ΔRecall	Acc.	ΔAcc.	Prec.	ΔPrec.	ROC-AUC	ΔROC-AUC	PR-AUC	ΔPR-AUC
P1	**No Balancing (baseline — *without*)**	0.2166	—	0.1354	—	0.8330	—	0.5416	—	0.7713	—	0.4010	—
P1	Random Oversampling	0.4435	+0.2269	0.7033	+0.5679	0.6990	–0.1340	0.3239	–0.2177	0.7715	+0.0002	0.3995	–0.0016
P1	SMOTE	0.4410	+0.2244	0.6951	+0.5597	0.6995	–0.1335	0.3230	–0.2186	0.7685	–0.0027	0.3974	–0.0036
P1	ADASYN	0.4359	+0.2193	0.7224	+0.5870	0.6812	–0.1518	0.3122	–0.2294	0.7669	–0.0044	0.3952	–0.0058
P2	Borderline-SMOTE	0.4399	+0.2233	0.7062	+0.5708	0.6933	–0.1397	0.3195	–0.2221	0.7686	–0.0027	0.3962	–0.0048
P2	KMeans-SMOTE	0.4189	+0.2023	0.4740	+0.3386	0.7757	–0.0573	0.3753	–0.1663	0.7569	–0.0144	0.3773	–0.0237
P2	SVM-SMOTE †	0.4398	+0.2231	0.5992	+0.4638	0.7396	–0.0934	0.3473	–0.1943	0.7631	–0.0082	0.3914	–0.0096
P2	SMOTE-N †	0.4201	+0.2034	0.6255	+0.4901	0.7054	–0.1276	0.3162	–0.2254	0.7374	–0.0338	0.3716	–0.0294
P3	Tomek Links	0.2242	+0.0075	0.1416	+0.0062	0.8329	–0.0001	0.5385	–0.0031	0.7713	+0.0001	0.4011	+0.0001
P3	ENN	0.4262	+0.2096	0.4467	+0.3113	0.7948	–0.0381	0.4075	–0.1341	0.7715	+0.0003	0.4006	–0.0004
P3	SMOTE-Tomek	0.4410	+0.2244	0.6943	+0.5589	0.6998	–0.1332	0.3232	–0.2184	0.7684	–0.0028	0.3973	–0.0037
P3	SMOTE-ENN	0.3191	+0.1024	0.2455	+0.1101	0.8213	–0.0117	0.4556	–0.0860	0.7526	–0.0187	0.3781	–0.0229

Table 2. Random Forest — baseline vs each technique (deltas)

P	Technique	F1	ΔF1	Recall	ΔRecall	Acc.	ΔAcc.	Prec.	ΔPrec.	ROC-AUC	ΔROC-AUC	PR-AUC	ΔPR-AUC
P1	**No Balancing (baseline — *without*)**	0.1531	—	0.0876	—	0.8347	—	0.6076	—	0.7775	—	0.4137	—
P1	Random Oversampling	0.4441	+0.2910	0.7385	+0.6509	0.6847	–0.1500	0.3176	–0.2900	0.7777	+0.0002	0.4137	0.0000
P1	SMOTE	0.4415	+0.2884	0.7315	+0.6438	0.6843	–0.1504	0.3162	–0.2914	0.7751	–0.0024	0.4084	–0.0053
P1	ADASYN	0.4321	+0.2790	0.7716	+0.6840	0.6541	–0.1806	0.3001	–0.3075	0.7727	–0.0048	0.4032	–0.0105
P2	Borderline-SMOTE	0.4361	+0.2830	0.7618	+0.6741	0.6640	–0.1707	0.3055	–0.3020	0.7740	–0.0036	0.4023	–0.0114
P2	KMeans-SMOTE	0.3918	+0.2386	0.3803	+0.2927	0.7986	–0.0361	0.4039	–0.2036	0.7635	–0.0140	0.3752	–0.0385
P2	SVM-SMOTE †	0.4405	+0.2874	0.5741	+0.4865	0.7512	–0.0835	0.3573	–0.2502	0.7676	–0.0099	0.3980	–0.0157
P2	SMOTE-N †	0.4130	+0.2598	0.5720	+0.4844	0.7226	–0.1121	0.3231	–0.2844	0.7383	–0.0392	0.3597	–0.0540
P3	Tomek Links	0.1636	+0.0104	0.0946	+0.0070	0.8349	+0.0002	0.6023	–0.0052	0.7776	+0.0001	0.4139	+0.0002
P3	ENN	0.4301	+0.2770	0.4401	+0.3524	0.8011	–0.0336	0.4207	–0.1869	0.7772	–0.0003	0.4127	–0.0010
P3	SMOTE-Tomek	0.4416	+0.2885	0.7319	+0.6443	0.6843	–0.1504	0.3162	–0.2914	0.7750	–0.0026	0.4083	–0.0054
P3	SMOTE-ENN	0.3264	+0.1733	0.2501	+0.1625	0.8239	–0.0108	0.4697	–0.1379	0.7425	–0.0351	0.3823	–0.0314

Table 3. XGBoost — baseline vs each technique (deltas)

P	Technique	F1	ΔF1	Recall	ΔRecall	Acc.	ΔAcc.	Prec.	ΔPrec.	ROC-AUC	ΔROC-AUC	PR-AUC	ΔPR-AUC
P1	**No Balancing (baseline — *without*)**	0.2034	—	0.1232	—	0.8354	—	0.5832	—	0.7802	—	0.4176	—
P1	Random Oversampling	0.4453	+0.2419	0.7463	+0.6231	0.6829	–0.1525	0.3173	–0.2659	0.7802	–0.0001	0.4176	0.0000
P1	SMOTE	0.4415	+0.2380	0.7321	+0.6088	0.6840	–0.1514	0.3160	–0.2672	0.7742	–0.0060	0.4079	–0.0098
P1	ADASYN	0.4303	+0.2269	0.7689	+0.6457	0.6528	–0.1827	0.2988	–0.2845	0.7703	–0.0100	0.3998	–0.0178
P2	Borderline-SMOTE	0.4367	+0.2333	0.7601	+0.6369	0.6655	–0.1699	0.3063	–0.2769	0.7734	–0.0068	0.3991	–0.0185
P2	KMeans-SMOTE	0.3986	+0.1951	0.3896	+0.2664	0.7995	–0.0360	0.4080	–0.1752	0.7701	–0.0101	0.3809	–0.0367
P2	SVM-SMOTE †	0.4389	+0.2355	0.5857	+0.4625	0.7446	–0.0909	0.3509	–0.2323	0.7646	–0.0156	0.3965	–0.0211
P2	SMOTE-N †	0.4002	+0.1968	0.5583	+0.4351	0.7146	–0.1209	0.3119	–0.2713	0.7265	–0.0538	0.3496	–0.0680
P3	Tomek Links	0.2140	+0.0106	0.1314	+0.0082	0.8354	0.0000	0.5769	–0.0063	0.7803	0.0000	0.4176	0.0000
P3	ENN	0.4420	+0.2386	0.4794	+0.3562	0.7935	–0.0419	0.4100	–0.1732	0.7799	–0.0003	0.4168	–0.0009
P3	SMOTE-Tomek	0.4414	+0.2380	0.7301	+0.6069	0.6848	–0.1506	0.3163	–0.2669	0.7740	–0.0062	0.4076	–0.0100
P3	SMOTE-ENN	0.3449	+0.1415	0.2797	+0.1565	0.8188	–0.0166	0.4500	–0.1333	0.7461	–0.0341	0.3829	–0.0347

Key takeaways

1. **F1 roughly doubles** with balancing: the largest gains are Random Oversampling (+0.227 LR, +0.291 RF, +0.242 XGB) — baseline F1 was only 0.153–0.217.
2. **Recall jumps dramatically**: ADASYN gains +0.587 (LR), +0.684 (RF), +0.646 (XGB) — the model goes from missing ~7 in 8 diabetic cases to catching ~3 in 4.
3. **Accuracy and precision fall** as the expected price: accuracy deltas run –0.04 to –0.18, precision –0.19 to –0.29. The baseline's 0.83–0.84 accuracy flattered a model that detected almost nothing.
4. **Ranking quality barely moves**: ROC-AUC deltas stay within ±0.04 (mostly ±0.01) and PR-AUC deltas within ±0.03 — for XGBoost, PR-AUC is identical (0.4176) with and without balancing. Resampling shifts the *operating point*, not the ranking.
5. **Tomek Links ≈ baseline** on every delta (all near zero): the gentlest cleaning changes almost nothing.
6. **ENN is the accuracy-preserving exception**: F1 +0.210–0.239 with accuracy deltas of only –0.034 to –0.042.

Caveats: † rows rest on a 50k training subsample; KMeans-SMOTE used non-default parameters; XGBoost differences below ~0.002 are cross-machine noise. Full detail: `P4_Integration_Evaluation/report/Person_4_Integration_Report.md` §4, §9.

Generated 2026-10-18 from the validated integrated results. Deltas computed programmatically; no hand-transcribed values.